

STAT 20000 Chapter 19

Sample Survey

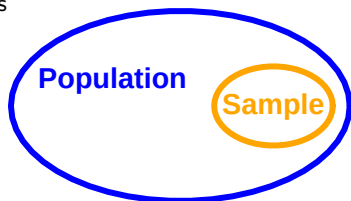
Four Keywords in Sampling

Population: The entire group of individuals in which we are interested but can't usually assess directly.

Example: All humans, all working-age people in California, all crickets

Sample: The part of the population we actually examine and for which we do have data.

How well the sample represents the population depends on the sample design.



A **parameter** is a number describing a characteristic of the population

A **statistic** is a number describing a characteristic of a sample.

Example

Suppose we want to predict the result of an election in a state.

A sample of 3000 registered voters are interviewed:

45% support candidate A, and 40% support candidate B.

► **population:**

Example

Suppose we want to predict the result of an election in a state.

A sample of 3000 registered voters are interviewed:

45% support candidate A, and 40% support candidate B.

- ▶ **population:** registered voters in the state that is going to vote
- ▶ **parameter:**

Example

Suppose we want to predict the result of an election in a state.

A sample of 3000 registered voters are interviewed:

45% support candidate A, and 40% support candidate B.

- ▶ **population:** registered voters in the state that is going to vote
- ▶ **parameter:** the percentage of votes for candidate A, and also the percentage of votes for candidate B
- ▶ **sample:**

Example

Suppose we want to predict the result of an election in a state.

A sample of 3000 registered voters are interviewed:

45% support candidate A, and 40% support candidate B.

- ▶ **population:** registered voters in the state that is going to vote
- ▶ **parameter:** the percentage of votes for candidate A, and also the percentage of votes for candidate B
- ▶ **sample:** the 3,000 registered voters interviewed
- ▶ **statistic:**

Example

Suppose we want to predict the result of an election in a state.

A sample of 3000 registered voters are interviewed:

45% support candidate A, and 40% support candidate B.

- ▶ **population**: registered voters in the state that is going to vote
- ▶ **parameter**: the percentage of votes for candidate A, and also the percentage of votes for candidate B
- ▶ **sample**: the 3,000 registered voters interviewed
- ▶ **statistic**: the percentage in the sample that support candidate A, and that for candidate B

Tale 1: The *Literary Digest* Poll

Literary Digest

- ▶ well-known magazine in the U.S in 1890-1936
- ▶ Old issues at Regenstein
- ▶ had run presidential polls since 1920; always correct
- ▶ went bankrupt in 1938

The 1936 election

- ▶ FDR vs. Alfred Landon
- ▶ *Literary Digest* mailed 10 million people (20% of voters in the country)
- ▶ Sample from phone lists, auto registrations, and club registers
- ▶ 2,376,523 postcard replies, response rate $\approx 24\%$

	FDR	Landon	Lemke	Sample Size
Literary Digest	41%	55%	4%	2.4 million
Gallup	56%	?	?	50,000
Election Result	61%	37%	2%	

FDR won in 46 states and Landon in 2 states (523 v.s. 8 electoral votes).

Why *Literary Digest* Poll Failed?

Mistake #1: in 1936, poor people were more likely not to have telephones or join clubs;

- ▶ Systematic tendency in the sampling procedure to exclude one kind of persons from the sample — **selection bias**

When a selection procedure is biased, taking a large sample does not help. This just repeats the basic mistake on a larger scale.

Why *Literary Digest* Poll Failed?

Mistake #1: in 1936, poor people were more likely not to have telephones or join clubs;

- ▶ Systematic tendency in the sampling procedure to exclude one kind of persons from the sample — **selection bias**

When a selection procedure is biased, taking a large sample does not help. This just repeats the basic mistake on a larger scale.

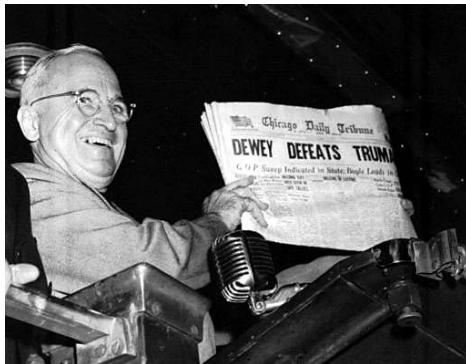
Mistake #2: Only 24% surveyed people responded

- ▶ Non-respondents can differ from respondents in many important ways;
- ▶ Ignoring non-respondents introduces **non-response bias**.

Non-respondents can be very different from respondents. When there is a high non-response rate, look out for non-response bias.

Tale 2: Election 1948, Truman vs. Dewey

<i>Candidates</i>	Predictions			<i>Results</i>
	<i>Crossley</i>	<i>Gallup</i>	<i>Roper</i>	
Truman	45	44	38	50
Dewey	50	50	53	45



Truman grinned and held up a copy of *Chicago Tribune* that erroneously announced his defeat

Quota Sampling

- ▶ Each interviewer in was assigned a fixed quota of subjects to interview;
- ▶ Certain categories of interest were fixed beforehand, e.g., sex, age, race, and economic status;
- ▶ The numbers of subjects falling into the categories were fixed;
- ▶ E.g., an interviewer St. Louis was required to interview 13 subjects, of whom
 - ▶ 6 lived in the suburbs, and 7 in the central city;
 - ▶ 7 were to be men, and 6 to be women;
Of the 7 men
 - ▶ 3 were to be under forty years old, and 4 over forty;
 - ▶ 1 was to be black, and 6 white,
and so on;
- ▶ In other respects, the interviewer were free to select anybody they liked

Quota Sampling (Cont'd)

- ▶ Improvement over the *Literary Digest* poll: clear intention to represent faithfully some key characteristics that affect voting behavior, for example, demographic distribution.

Quota Sampling (Cont'd)

- ▶ Improvement over the *Literary Digest* poll: clear intention to represent faithfully some key characteristics that affect voting behavior, for example, demographic distribution.
- ▶ Problem #1: There are many factors which influence voting behavior besides the ones the survey organization control for.

Quota Sampling (Cont'd)

- ▶ Improvement over the *Literary Digest* poll: clear intention to represent faithfully some key characteristics that affect voting behavior, for example, demographic distribution.
- ▶ Problem #1: There are many factors which influence voting behavior besides the ones the survey organization control for.
- ▶ Problem #2: The subjects are **hand-picked**.
The interviewers unintentionally picked more Republicans than Democrats because historically Republicans were
 - ▶ more likely to have a phone,
 - ▶ more likely to have a permanent home,
 - ▶ more likely to live in a nice neighborhood

It was also the case in 1948 election

Some Bad Sampling Methods

- ▶ **Convenience Sampling** — Just ask whoever is around.
 - ▶ e.g., man-on-the-street survey
 - ▶ drawback: results may vary greatly with “when and where” the survey is done, lack of representation

Some Bad Sampling Methods

- ▶ **Convenience Sampling** — Just ask whoever is around.
 - ▶ e.g., man-on-the-street survey
 - ▶ drawback: results may vary greatly with “when and where” the survey is done, lack of representation
- ▶ **Voluntary Response Sampling**
 - ▶ e.g., internet polls, call-in surveys
 - ▶ drawback 1: selection bias — only those visiting the website/watching the program are sampled
 - ▶ drawback 2: those with extreme opinions are more likely to participate

Some Bad Sampling Methods

- ▶ **Convenience Sampling** — Just ask whoever is around.
 - ▶ e.g., man-on-the-street survey
 - ▶ drawback: results may vary greatly with “when and where” the survey is done, lack of representation
- ▶ **Voluntary Response Sampling**
 - ▶ e.g., internet polls, call-in surveys
 - ▶ drawback 1: selection bias — only those visiting the website/watching the program are sampled
 - ▶ drawback 2: those with extreme opinions are more likely to participate

Better Sampling Methods

- ▶ Simple Random Sampling
- ▶ Multistage Clustered Sampling
- ▶ ⋮

Simple Random Sampling (SRS)

Basic idea: put the names in a box and make draws from the box

- ▶ need a list of names of all subjects in the population, called **sampling frame**
- ▶ all subjects have the same chance to be chosen
- ▶ the Law of Large Number ensures the makeup of a SRS to mimic the makeup of the population (age/gender/race/income...)
- ▶ impractical for large population

Multistage Cluster Sampling

- ▶ First stage: the population is divided into groups, called **clusters**, and a sample of groups is chosen.
- ▶ Second stage: the selected groups is further divided into subgroups, and a sample of subgroups is chosen in each selected group
- ▶ (Third stage: ...)
- ▶ (Fourth stage: ...)

Multistage Cluster Sampling

- ▶ First stage: the population is divided into groups, called **clusters**, and a sample of groups is chosen.
- ▶ Second stage: the selected groups is further divided into subgroups, and a sample of subgroups is chosen in each selected group
- ▶ (Third stage: ...)
- ▶ (Fourth stage: ...)

In 1952-1984, Gallop used this method in pre-election survey.

- ▶ towns → wards → precincts → households (FPP, p.341)

Multistage Cluster Sampling

- ▶ First stage: the population is divided into groups, called **clusters**, and a sample of groups is chosen.
- ▶ Second stage: the selected groups is further divided into subgroups, and a sample of subgroups is chosen in each selected group
- ▶ (Third stage: ...)
- ▶ (Fourth stage: ...)

In 1952-1984, Gallop used this method in pre-election survey.

- ▶ towns → wards → precincts → households (FPP, p.341)

Advantage:

- ▶ less traveling cost to send interviewers around,
- ▶ no need to make sampling frame for unselected subgroups

Other Common Problems in Sample Surveys

- ▶ **Undercoverage** – part of the population got left out of the sampling frame
- ▶ **Non-response bias** – discussed already
 - ▶ Remedies: call back, double sampling scheme
- ▶ **Response bias** – the answers by respondents are influenced to some extent by the phrasing of the questions, and even the tone or attitude of the interviewer.
 - ▶ Remedies: interviewer control, proper design of questionnaires
- ▶ Check data – put less weight on a group if the sample includes proportionately too many respondents from the group